

Highlights

Surrogate Learning of Scour-critical Bridge Capacities and Uncertainty Quantification Considering Climate Scenarios

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- A distribution-agnostic framework is proposed to assess scour-critical bridge capacity under climate-driven deep uncertainty.
- Surrogate models are trained using nonlinear pushover simulations with climate-scenario-based scour inputs.
- Credal-set uncertainty quantification is introduced to provide bounded predictions, ensuring objective capacity estimation.
- Results support resilient decision-making for infrastructure adaptation and asset management under uncertain climate futures.

Surrogate Learning of Scour-critical Bridge Capacities and Uncertainty Quantification Considering Climate Scenarios

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Abstract

Scour-critical bridges are among the most vulnerable transportation infrastructure systems, and their failure risk is expected to be increasingly uncertain due to climate change and climate-driven flood hazards. Traditional approaches often overlook the effects of such deep uncertainties and compute scour hazards assuming a single probabilistic distribution to analyze bridge capacity and collapse potential. This study develops a distribution-agnostic framework to predict the transverse capacities and quantify the uncertainties of scour-critical bridges. The major components include: (1) surrogate modeling of bridge capacities through learning from nonlinear pushover-based simulated data considering climate-scenario-based scour profiles, and (2) credal-set quantification of capacity uncertainties subject to climate-scenario scour profiles. The results reveal how variable uncertainties propagate through the nonlinear soil–foundation–structure bridge system when

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expressing the capacity tuple, including base shear, deck displacement, base moment, and column rotation. The resulting framework provides a novel analytics toolkit for climate-change and uncertainty-aware assessment of scour-critical bridge capacity, supporting engineers and transportation agencies in making resilient asset management decisions.

Keywords: Scour, Bridge Capacity, Surrogate Modeling, Climate Scenario, Credal Set, Soil–foundation–structure Interaction,

1. Introduction

Waterway bridge structures serve as critical nodes in transportation infrastructure, enabling vital socioeconomic connectivity. However, they are especially vulnerable to scour: the erosion of soil surrounding bridge foundations caused by flowing water. In the United States, over 600,000 bridges span the nation’s waterways, and an estimated 70% were not originally designed to resist scour. Of these, approximately 21,000 are classified as *scour-critical* (USGS, 2018). Scour has historically been the leading cause of highway bridge failure, accounting for more than half of such collapses nationwide (Simpson and Byun, 2019). Looking ahead, climate change is projected to intensify scour risk in the 21st century, as more frequent and severe riverine floods accelerate foundation erosion (Cranston and Smith, 2022). This convergence of inadequate original design, material aging, and worsening hydrologic hazards due to climate change highlights the need for predictive, risk-informed assessments of scour-critical bridges throughout their service life.

Developing reliable life-cycle performance quantification for scour-critical

18 bridges is technically challenged by numerous factors. Among these, bridge
19 scour as a hazard differs fundamentally from other natural and technological
20 hazards (e.g., earthquakes and barge impacts). One key distinction lies in
21 the uncertainties associated with scenario-based climate projections that are
22 inherent in mainstream climate modeling methodologies, whether based on
23 global climate models or downscaled regional models (Alizadeh, 2022; O'Neill
24 et al., 2020). Researchers have long referred to such uncertainties as deep
25 uncertainties (DUs), attributing them to hard-to-quantify or unknown fac-
26 tors such as long-term socioeconomic development, emission policy decisions,
27 and technological change (Kandlikar et al., 2005). The Intergovernmental
28 Panel on Climate Change (IPCC) addresses these uncertainties by employ-
29 ing multiple climate scenario frameworks (e.g., Representative Concentration
30 Pathways) that span optimistic to pessimistic emission trajectories. When
31 adopted in hazard modeling (e.g., precipitation), these climate-driven DUs
32 propagate through regional hydrological and hydraulic models (e.g., basin
33 and riverine floods), and ultimately influence local scour profiles at water-
34 way bridges.

35 Although researchers have acknowledged the complexity and implications
36 of climate scenarios and deep uncertainties (DUs) for civil infrastructure de-
37 sign and management (Helmrich and Chester, 2022), few studies have explic-
38 itly examined how these uncertainties affect bridge safety or design. Only
39 in recent years have efforts emerged to integrate climate model outputs into
40 scour risk assessments. For instance, Yang et al. linked downscaled climate
41 data to bridge failure probabilities (Yang and Frangopol, 2019), and addi-
42 tional relevant studies are reviewed below. Nevertheless, a systematic treat-

43 ment of scenario-driven uncertainty in bridge performance remains limited.
44 Several investigations have incorporated climate-scenario information into
45 scour or bridge risk modeling, yet significant gaps persist. Khelifa et al.
46 quantified the influence of intensified flood regimes on scour-failure probabil-
47 ities and economic losses; however, their framework employed deterministic
48 structural limits and did not propagate climate-projection uncertainty into
49 structural capacity metrics (Khelifa et al., 2013). Dikanski et al. explored
50 the interaction between climate-model variability and asset-data uncertainty
51 for a U.K. rail-bridge inventory, revealing that inadequate inventory data
52 can overshadow climate uncertainty by an order of magnitude. Still, their
53 analysis focused on the network level and did not address component-level
54 performance or surrogate modeling (Dikanski et al., 2018). Liu et al. de-
55 veloped a network-level, risk-based adaptation framework for bridges facing
56 climate-aggravated scour, but their approach relied on computationally ex-
57 pensive Monte Carlo simulations and was limited to discrete scenarios (Liu
58 et al., 2020). Nasr et al. generated a national-scale map of Swedish catch-
59 ments vulnerable to future pier scour but did not translate increased scour
60 depths into structural safety margins (Nasr et al., 2023). Sasidharan et al.
61 proposed a “warning-time-to-failure” index for climate-informed scour man-
62 agement while omitting probabilistic modeling of structural capacity or yield-
63 based metrics (Sasidharan et al., 2023). Collectively, these studies highlight
64 the need to model the systems-level behavior and scour risk considering cli-
65 mate scenarios; nonetheless, they do not focus on the intrinsic properties
66 (namely, capacities) of scour-critical bridges.

67 In Bridge Engineering practice, load rating has been a traditional ap-

68 proach for evaluating the vertical capacity of in-service bridges (Cai and
69 Shahawy, 2003; Scott et al., 2008; Gokce et al., 2011). This procedure typi-
70 cally involves structural analysis based on LRFD-prescribed combinations of
71 dead and live loads, hence defining the vertical capacity of bridges. For the
72 assessment of waterway bridge collapse potential, the lateral capacity rating
73 has primarily relied on nonlinear modal pushover analysis (Alipour et al.,
74 2013; Bignell et al., 2005). These methods yield nonlinear pushover curves
75 that are essential for characterizing lateral capacity under extreme transverse
76 loads, such as those induced by seismic, hydraulic, or barge impact events.
77 Capacity estimates are typically defined at the point of yielding, represented
78 by the corresponding load and displacement values. However, few studies
79 have addressed the variation in structural capacity as a function of vary-
80 ing scour depth. Moreover, no existing studies have explicitly incorporated
81 climate scenario-induced scour uncertainties. A key technical barrier in ad-
82 dressing these gaps is the high computational cost associated with modeling
83 complex nonlinear bridge systems under variable scour conditions.

84 Given the aforementioned research gaps, there is a compelling need to
85 integrate modern data-driven modeling approaches for modeling and under-
86 standing lateral capacities of scour-critical bridges with the deep uncertainties
87 inherent in climate scenarios. These machine learning (ML) methods, while
88 distinct from traditional finite element or first-principles-based approaches,
89 often remain dependent on simulation data generated through conventional
90 techniques. However, once trained on such data, the resulting ML models, i.e.
91 surrogates, can serve as computationally efficient predictive tools. Moreover,
92 by directly mapping inputs (e.g., scour depth) to outputs (e.g., structural

capacities), these models enable efficient quantification of predictive uncertainty—an essential capability when addressing scenario-based deep uncertainties central to this study.

This study proposes a novel modeling framework, which includes (1) surrogate learning of the structural capacities of scour-critical bridges under climate scenario-induced deep uncertainties and (2) quantification of climate-scenario-induced scour uncertainties without assuming a specific distribution of scour profiles. Statistically, the proposed framework is distribution-agnostic in nature. In summary, the paper presents the following key contributions:

1. *Surrogate Learning Framework for Scour Capacity.* A surrogate learning framework is developed to efficiently predict the structural capacity of scour-affected bridges using a library of nonlinear pushover analyses, enabling rapid capacity estimation without the need for repeated high-fidelity simulations.
2. *Capacity Tuple Definition.* A multi-dimensional capacity tuple is introduced to capture critical performance measures at yielding, including: (i) base shear, (ii) deck displacement, (iii) pile-cap moment, and (iv) pile-cap rotation—collectively characterizing both superstructure and substructure limit states.
3. *Climate-Induced Deep Uncertainty Quantification.* Deep uncertainties are incorporated via a credal-set approach, providing upper and lower bounds on capacity predictions across various climate scenario trajectories and yielding a robust envelope of predictive outcomes.

In the following, this paper starts with a literature review of related efforts

118 and techniques used in this work. Subsequently, it presents the proposed
119 surrogate learning and uncertainty quantification framework. The numerical
120 results given the select climate-scenario scour hazards and a representative
121 bridge model are presented. The paper further discusses the implications,
122 limitations, and directions for future research, followed by the conclusion.

123 **2. Related Efforts and Techniques**

124 *2.1. Related Efforts*

125 Numerous studies have addressed the impact of variable scour profiles
126 within fragility modeling frameworks. In some approaches, scour depth is
127 treated as a conditional variable; in others, a fully probabilistic treatment
128 is adopted, modeling scour as a hazard with an assigned distribution. Since
129 scour not only acts as a primary hazard but also alters the boundary condi-
130 tions at the foundation level, it is frequently analyzed in combination with
131 other extrinsic natural hazards, resulting in multi-hazard assessment frame-
132 works. Within this domain, several key findings have emerged:

- 133 • *Scour + Seismic.* Wang et al. (Wang et al., 2014) showed that even
134 moderate scour depths (< 2 m) can double the probability of column
135 yielding during design-level earthquakes. Guo and Chen (Guo and
136 Chen, 2015) extended this work to a life-cycle framework, while Guo et
137 al. (Guo et al., 2016) developed time-dependent fragility functions that
138 shift toward lower peak ground accelerations (PGAs) as scour severity
139 increases. Parametric and copula-based approaches (e.g., Argyroudis
140 and Mitoulis (Argyroudis and Mitoulis, 2021)) corroborate these find-
141 ings across various bridge typologies.

- 142 • *Scour + Barge Impact.* Kameshwar and Padgett (Kameshwar and Pad-
143 gett, 2018) incorporated scour-dependent lateral support loss into ves-
144 sel collision models, finding that collapse probability increases sharply
145 once scour depth exceeds roughly 1.5 pier diameters. Fan et al. further
146 integrated corrosion effects and nonlinear soil–pile interaction, reveal-
147 ing a shift in dominant failure mode from the superstructure to the
148 pile foundation (Fan et al., 2021). Guo et al. (2023) developed impact
149 fragility analysis to characterize bridge failure under barge collision and
150 local scour hazards.
- 151 • *Scour + Flood Debris / Flow.* Anisha et al. (Anisha et al., 2022)
152 jointly modeled debris impact, inundation pressure, and scour, con-
153 cluding that a 1.5 m scour depth increases flood-induced failure prob-
154 ability by approximately 80%. Karriqi et al. (Karriqi et al., 2024)
155 investigated sequential loading scenarios (earthquake followed by flood-
156 induced scour), observing moderate but systematic increases in system-
157 level fragility.

158 Despite variations in modeling strategies, these studies share a common
159 approach to probabilistically estimating structural collapse. For a generic
160 structure, this begins by defining a displacement-based capacity measure:
161 typically, the yielding displacement denoted Δ_y , obtained via nonlinear pushover
162 analysis. A corresponding response-demand quantity, Δ , is then computed
163 by simulating structural response under hazardous forces (e.g., from seismic
164 or barge-impact loadings) characterized by an intensity measure H . The col-
165 lapse event is then defined as the normalized demand $u = \Delta/\Delta_y$ exceeding a

166 threshold value u_t , namely $u > u_t$. Through stochastic simulation that incor-
 167 porates load and material uncertainties, researchers estimate the conditional
 168 probability of collapse, $P(u > u_t | H)$, namely collapse fragility. For conven-
 169 tional structures with fixed boundary conditions, Δ_y is treated as a constant
 170 property of the structure. However, in scour-critical bridges, Δ_y varies with
 171 scour depth, as scour alters the boundary conditions at the foundation level.
 172 Furthermore, it introduces ambiguity in selecting an appropriate threshold
 173 constant, u_t .

174 To illustrate this concept, Figure 1 presents three nonlinear pushover
 175 profiles corresponding to different scour conditions. The solid curve, with
 176 yielding capacity (Δ_{y0}, V_{y0}) , represents the non-scour case. The dashed curve
 177 with (Δ_{y1}, V_{y1}) corresponds to a moderate scour scenario, and the dot-dashed
 178 curve with (Δ_{y2}, V_{y2}) reflects a severe scour condition. If a constant capacity,
 179 such as Δ_{y0} , is used to evaluate collapse potential across all scenarios, the
 180 resulting index $u = \Delta/\Delta_{y0}$ would misleadingly suggest equal vulnerability
 181 to collapse for the underlying bridge with the varying scour conditions. This
 182 highlights a fundamental limitation in using fixed capacity metrics under
 183 variable scour conditions. Moreover, as shown in Figure 1, both displace-
 184 ment and strength capacities (Δ_y and V_y) are reduced with increasing scour
 185 severity. Consequently, collapse limit states for scour-critical bridges must
 186 account for reductions in both strength and displacement capacities to reflect
 187 realistically the system capacity under scour-induced degradation.

188 2.2. Climate-scenario Uncertainty and Spatial Analogue

189 Decision-making in hydraulic and infrastructure engineering under cli-
 190 mate change is increasingly challenged by *deep uncertainties* associated with

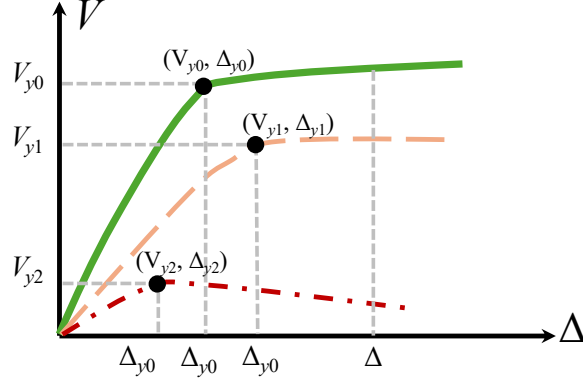


Figure 1: Illustration of representative nonlinear pushover curves for scour-critical bridge structures under varying scour profiles.

191 various climate scenarios (IPCC, 2022). This condition is particularly acute
 192 in the context of projecting localized flood hazards, where uncertainty propa-
 193 gates across multiple modeling layers, from global climate models (GCMs) to
 194 regional downscaling, and more localized hydrological simulations (Lafferty
 195 and Sriver, 2023).

196 Under such uncertainty, conventional risk-based planning methods that
 197 rely on probabilistic forecasts become insufficient. As emphasized by Lem-
 198 pert et al. (Lempert et al., 2024), deep uncertainty necessitates alternative
 199 strategies that explore a broad range of plausible futures rather than opti-
 200 mizing for a single forecast. In this context, *Robust Decision Making (RDM)*
 201 and *Dynamic Adaptive Policy Pathways (DAPP)* have emerged as leading
 202 frameworks.

203 One practical implementation of these strategies in hydrological risk as-
 204 sessment is the use of **spatial analogue**: the substitution of geographically
 205 distinct but hydrologically relevant locations to represent potential future

206 conditions at a target site. This space-for-time reasoning is rooted in the
 207 ergodic hypothesis, which asserts that spatial ensembles governed by simi-
 208 lar physical processes can approximate temporal evolution (Sivapalan et al.,
 209 2011). In hydrology, this principle supports the idea that rivers in disparate
 210 climate regimes today may serve as empirical analogues for the future be-
 211 havior of a local watershed under climate change.

212 This analogue-based scenario selection offers key advantages: it anchors
 213 scenario construction in observed extremes, supports empirical stress-testing
 214 aligned with IPCC guidance (IPCC, 2022), and enables site-specific yet trans-
 215 ferable hazard modeling. In this study, selected watersheds serve as proxies
 216 for future conditions, forming the empirical basis for generating synthetic
 217 but physically plausible flood and scour scenarios. Moreover, by bypass-
 218 ing reliance on predefined probabilistic distributions, the spatial-analogue
 219 approach directly supports a distribution-agnostic framework for modeling
 220 scour-critical bridge capacities under deep climate uncertainty.

221 *2.3. Surrogate Modeling Methods*

222 Data-driven surrogate models have emerged as efficient alternatives for
 223 simulating nonlinear structural responses under uncertainty. Among them,
 224 Support Vector Regression (SVR) and Gradient Boosting Regression (GBR)
 225 are widely used due to their complementary strengths in learning complex re-
 226 lationships from limited or structured data. SVR aims to find a function $f(\mathbf{x})$
 227 that minimizes a regularized ϵ -insensitive loss, promoting both model flatness
 228 and tolerance to small errors. This makes SVR particularly effective when
 229 training datasets are small or the underlying structural behavior is smooth.
 230 GBR, on the other hand, constructs an additive ensemble of shallow deci-

231 sion trees, where each successive tree fits the residuals of the previous one.
 232 This sequential learning strategy enables GBR to capture sharp nonlinear
 233 transitions and feature interactions effectively. Comparatively, toward gen-
 234 eralization, SVR leverages kernel-based representations to generalize well in
 235 low-data regimes, while GBR achieves high predictive accuracy by exploiting
 236 structural hierarchies in unseen data. For formal mathematical formulations
 237 of SVR and GBR, readers are referred to (Smola and Schölkopf, 2004; Fried-
 238 man, 2001).

239 Many researchers adopted SVR and GBR as surrogate modeling tech-
 240 niques modeling structural behavior in Civil Structural and Infrastructure
 241 Engineering [e.g., (Xu et al., 2020; Wang et al., 2022)]. Specifically, recent
 242 benchmarking studies, including Zhao et al. (Zhao et al., 2023), demonstrate
 243 that SVR and GBR consistently outperform linear models and often rival
 244 deep learning approaches, while requiring significantly less computational
 245 effort. These properties are especially advantageous in uncertainty quantifi-
 246 cation tasks, where repeated evaluation of surrogate models is necessary. In
 247 this study, SVR and GBR were selected as surrogate modeling techniques
 248 based on their proven performance in structural engineering applications.

249 *2.4. Uncertainty Quantification Methods*

250 Modern machine learning and statistics provide several methods to esti-
 251 mate prediction uncertainty; representative ones include:

- 252 • *Bayesian Predictive Intervals:* Bayesian approaches (e.g., Gaussian
 253 processes) yield posterior distributions and credible intervals by treat-
 254 ing model parameters as random variables (Rasmussen and Williams,
 255 2006).

- 256 • *Conformal Prediction*: Conformal methods produce intervals with fi-
 257 nite sample coverage guarantees, assuming future data follow the same
 258 distribution as training Angelopoulos and Bates (2021)
- 259 • *Credal Sets*: Rather than a single distribution, a credal set represents
 260 a set of plausible probability measures for the output, capturing both
 261 aleatory and epistemic uncertainty (Walley, 1991; Höffman et al., 2022).

262 While Bayesian and conformal intervals perform well in many practical
 263 situations, they often underestimate uncertainty when facing distributional
 264 shifts or deep epistemic uncertainty. In this work, we adopt credal set meth-
 265 ods to characterize uncertainties associated with bridge capacity prediction.
 266 This nonparametric distribution-agnostic approach yields a robust envelope
 267 that widens in regions of high epistemic uncertainty, as model disagreement
 268 increases (Breiman, 1996).

269 3. Methodology

270 3.1. Overview

271 Figure 2 outlines the overall framework, which integrates three major
 272 components:

- 273 1. Scenario Selection, Scour Hazard, and FE Model Generation. This is a
 274 preparation stage that focuses on establishing the foundational inputs
 275 for simulation, including specific hydraulic flow scenarios and a suite
 276 of scour conditions generated using probabilistic sampling. The finite-
 277 element model of the bridge of interest is systematically modified for
 278 each sampled scour condition.

- 279 2. Nonlinear Pushover Simulation and Capacity Extraction. This com-
 280 ponent executes nonlinear analyses for each scour case and extracts
 281 yield-level capacity responses. The outputs form a dataset for surro-
 282 gate model training.
- 283 3. Surrogate Modeling and Credal-Set Uncertainty Quantification (UQ).
 284 The final stage replaces computationally expensive simulations with
 285 lightweight machine learning models, enabling rapid capacity prediction
 286 and uncertainty quantification.

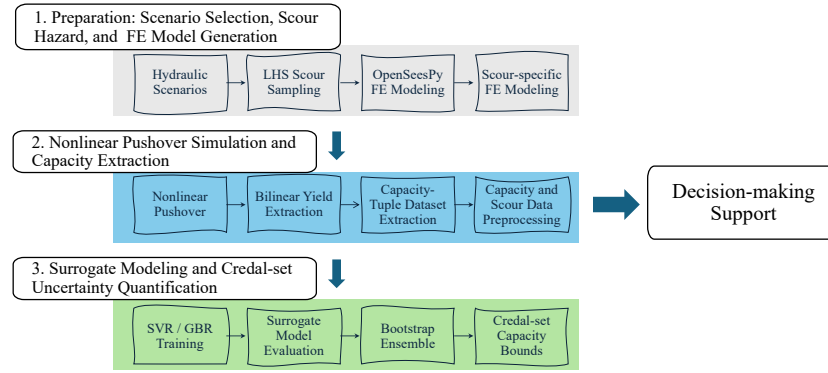


Figure 2: Proposed surrogate modeling and uncertainty quantification framework and methodological components.

287 Critical technical details and intermediate results from the climate-scenario-
 288 based scour analysis, FE modeling, and capacity value extraction are pre-
 289 sented in the following subsections. Technical methods for surrogate mod-
 290 eling and credal-set uncertainty quantification are described subsequently,
 291 followed by major result interpretations. The workflow is implemented using
 292 Python, including OpenSeesPy (Zhu, 2025) for FE simulation and Scikit-
 293 learn (scikit-learn Developers, 2025) for machine learning and uncertainty

294 quantification.

295 3.2. Scenario Selection and Scour Hazard Calculation

296 The specific design case involves a bridge spanning an imagined Midwest-
297 ern river channel. Given the difficulty of downscaling peak flow velocities
298 from global climate projections, the spatial-analogue concept is adopted in
299 this work. Each scenario corresponds to a flooding event of increasing sever-
300 ity and is tied to a potential climate future near the year 2050, as structured
301 by the SSP scenarios in IPCC AR6 (IPCC, 2022).

302 We identify two real-world rivers and one assumed worst-case as analogs
303 to capture the range of plausible future extremes (Table 1):

- 304 • *Scenario 1 – Moderate ($U = 2.9$ m/s)*: Based on average flood velocities
305 reported for the *Missouri River*, this case represents a baseline future
306 in which climate change causes moderate increases in flow velocities
307 (MDOT, 2019).
- 308 • *Scenario 2 – High ($U = 6.5$ m/s)*: Represents peak flood conditions in a
309 more extreme but still physically observed context, based on field mea-
310 surements from the *Colorado River* during a major flood event (Dawson
311 and Brunner, 2009). This scenario reflects the increased intensity and
312 flashiness expected in some AR6 pathways (e.g., SSP5-8.5).
- 313 • *Scenario 3 – Extreme ($U = 10$ m/s)*: A worst-case design flood veloc-
314 ity estimated for major Midwestern river systems under 500-year re-
315 currence conditions. This provides an upper envelope for stress-testing
316 structural safety and resilience.

317 To characterize long-term pier scour under variable hydraulic conditions,
318 this study adopts the erosion framework from Briaud et al. (Briaud et al.,
319 1999), widely used for cohesive soils and implemented in HEC-18. The
320 adopted erosion rates $\dot{z}$, classified based on Briaud’s framework, are sum-
321 marized in Table 1, corresponding to the selected scenarios and expected
322 hydraulic conditions. It is noted that herein the notation z is adopted for
323 scour and erosion following Briaud et al. (1999), which does not refer to the z
324 coordinate. Later, for surrogate modeling, S_z denotes the underlying random
325 variable for scour depth and s_z denotes a realized scour-depth value.

Table 1: Scenario-Based Hydraulic Inputs and Scour Predictions (LHS, $N = 1000$)

<i>Scenario</i>	U (m/s)	<i>Analog River (Flow Description)</i>	$\dot{z}$ (mm/h)	μ_{scour} (m)	σ (m)	95 th -Perc (m)
1	2.9	Missouri River (avg. flood)	100	2.68	1.13	4.81
2	6.5	Colorado River (2009 peak)	500	4.87	2.06	8.75
3	10.0	Midwestern extreme (500-yr design flood)	1000	6.44	2.72	11.56

326 The scour statistics in Table 1 are computed using the adopted bridge
327 pier diameter, $d_p = 1.5$ m, and 1,000 LHS samples for each scenario.

328 First, the maximum hydraulic shear stress τ_{max} is defined as a function of
329 upstream velocity U , water density ρ_w , and Reynolds number Re , computed
330 as:

$$Re = \frac{U d_p}{\gamma} \quad (1)$$

331 where d_p is the pier diameter and γ is the kinematic viscosity. The shear
332 stress τ_{max} is then given by the semi-empirical relation:

$$\tau_{\text{max}} = 0.094 \cdot \rho_w \cdot U^2 \left[\frac{1}{\log_{10}(Re)} - 0.1 \right] \quad (2)$$

333 The maximum scour depth under clear-water conditions is expressed as:

$$z_{\max} = 0.18 \cdot Re^{0.635} \quad (3)$$

334 This expression is derived from regression analysis of controlled tests con-
 335 ducted in cohesive soils (Gudavalli, 1998).

336 The equilibrium scour time t_{eq} is modeled as a duration-based erosion
 337 process that scales with time t , flow velocity U , and initial scour rate $\dot{z}$:

$$t_{\text{eq}} = 73 \cdot t^{0.126} \cdot U^{1.706} \cdot \dot{z}^{-0.2} \quad (4)$$

338 From this, the projected scour depth after 50 years, denoted z_{50} , is esti-
 339 mated by:

$$z_{50} = \frac{t_{\text{eq}}}{1/\dot{z} + t_{\text{eq}}/z_{\max}} \quad (5)$$

340 To capture uncertainty in long-term scour development due to stochastic
 341 variability in material properties and hydraulic conditions, a lognormal model
 342 is used for the final scour depth. Given a standard normal variate $\epsilon \sim \mathcal{N}(0, 1)$,
 343 the final scour depth is computed as:

$$z_{50}^{\text{final}} = \frac{e^{-0.085} \cdot z_{50} \cdot e^{0.407\epsilon}}{1000} \quad (6)$$

344 where the denominator converts millimeters to meters. This probabilistic
 345 formulation aligns with previous risk-informed studies on scour and bridge
 346 resilience (Alipour et al., 2013; Briaud and Oh, 2010).

347 To approximate the distribution of introduced random variables, the
 348 Latin Hypercube Sampling (LHS) method is adopted. The scour-depth sam-
 349 ples under the three different river flow conditions (Table 1) are simulated
 350 separately. For each scenario, the script generates 1,000 samples by convert-
 351 ing flow velocity into estimated scour depth. It accounts for uncertainty in

both hydraulic forces and soil erosion characteristics. Figure 3a and 3b illustrate the resulting distributions, including histogram plots of realized samples s_z , lognormal fits denoted by $f_{S_z}(z)$, and the fitted exceedance probability curves representing the hazard function: $P(S_z > z) = \int_z^\infty f_{S_z}(z') dz'$.

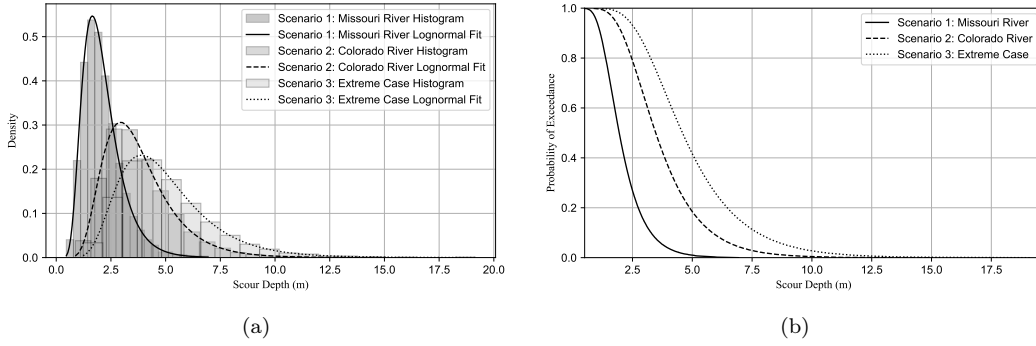


Figure 3: (a) Histogram distributions and lognormal fits for simulated climate scenario-based scour depth data. (b) Fitted hazard curves for cumulative distribution of simulated climate scenario-based scour depth data.

3.3. Finite-Element Modeling

As reviewed previously, many scour-critical bridge models have been developed in the literature. The framework proposed in this work is adaptable to any waterway bridge model, provided that climate-sensitive scour hazards are of interest. The bridge model adopted here is based on the one developed in Atak et al. (2014), originally implemented using the OpenSees framework with the TCL language. In this study, the model is re-implemented in Python using the OpenSeesPy library. Figure 4a illustrates the transverse elevation of the bridge structure, and Figure 4b provides details of the primary elements developed in the OpenSeesPy environment.

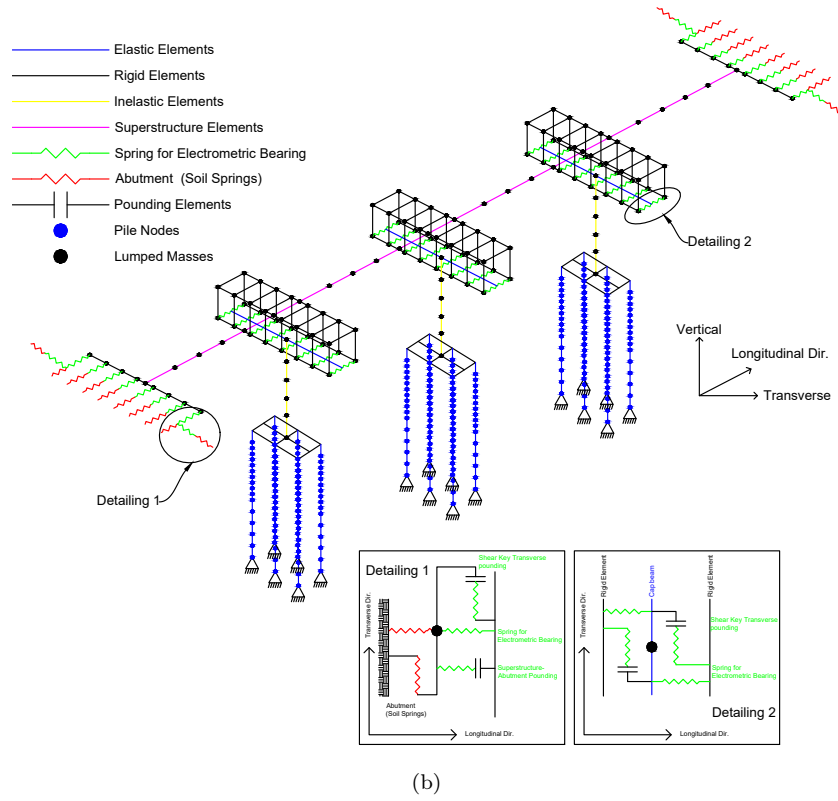
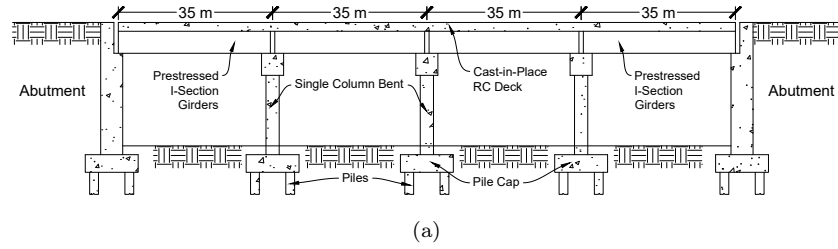


Figure 4: (a) Elevation view of the bridge structure. (b) The finite-element model with key nonlinear elements.

366 The modeled bridge represents a typical Midwestern U.S. bridge: a four-
 367 span, non-skewed prestressed concrete structure, with each span measuring
 368 35 meters and a total deck width of 12 meters. The superstructure consists
 369 of I-girders supporting a cast-in-place concrete deck, resting on elastomeric
 370 bearings over reinforced concrete piers and pile group foundations embed-
 371 ded in layered soil. In the model coordinate system, the original riverbed
 372 elevation is $z = -13.05$ m and the pile tips are located at $z = -33.05$ m;
 373 therefore, the modeled embedded pile length below the original riverbed is
 374 20 m. In the following analysis, s_z denotes a realized scour depth measured
 375 downward from the original riverbed, not the global z coordinate. The cor-
 376 responding scoured-bed elevation is $z_b = -13.05 \text{ m} - s_z$; for example, $s_z = 1$
 377 m corresponds to $z_b = -14.05$ m.

378 The piers are modeled using nonlinear beam-column elements with fiber
 379 sections to accurately capture flexural behavior, confinement, and post-yield
 380 response. Each pier column is divided into five integration points, provid-
 381 ing a detailed curvature distribution. The `Concrete01` material is used for
 382 both unconfined and confined concrete zones, while `Steel02` is adopted for
 383 reinforcing steel, with cyclic hardening parameters $R_0 = 15.0$, $c_{R1} = 0.9$,
 384 and $c_{R2} = 0.2$. Shear and torsional effects are captured using Aggregator
 385 sections.

386 Soil-structure interaction is modeled through nonlinear `zeroLength` springs
 387 at the pile-soil interface, simulating p - y lateral response and t - z vertical shaft
 388 response following FHWA HEC-18 recommendations. The lateral springs use
 389 depth-dependent `PySimple1` materials in the two horizontal directions, while
 390 the vertical shaft springs use depth-dependent `TzSimple1` materials in the

vertical direction. These p - y and t - z springs are assigned to all 18 modeled piles across the left, center, and right pile groups at common soil-spring elevations from $z = -13.05$ m to $z = -32.05$ m: the spacing is 0.5 m from $z = -13.05$ m to $z = -23.05$ m and 1.0 m from $z = -23.05$ m to $z = -32.05$ m. Springs at or above the scoured-bed elevation z_b are deactivated to simulate soil loss, thereby reducing lateral and axial stiffness. This treatment assumes a uniform scour depth across all pile groups; in an actual river channel, local hydraulics may produce non-uniform scour profiles among bents and individual piles, which can influence bridge response and is not explicitly modeled here (Ateş and Avşar, 2024).

To reflect scour-induced soil loss, springs are progressively removed as s_z increases. While material property uncertainties are included, they are intentionally minimized so that scour remains the dominant source of variability. Specifically, concrete compressive strength f'_c is modeled as a normal distribution with a mean of 27.0 MPa and a standard deviation of 1.0 MPa. Steel yield strength f_y is modeled using a lognormal distribution with a mean of 420.0 MPa and a standard deviation of 4.2 MPa.

Latin Hypercube Sampling (LHS) is employed with 1,000 samples per scenario (three scenarios in total), resulting in 3,000 simulations. This sampling approach robustly approximates the probabilistic scour hazard while keeping material variability constrained (Hwang et al., 2001). Figures 5a and 5b depict the resulting histograms based on the sampled values for concrete and reinforcing steel strength, respectively.

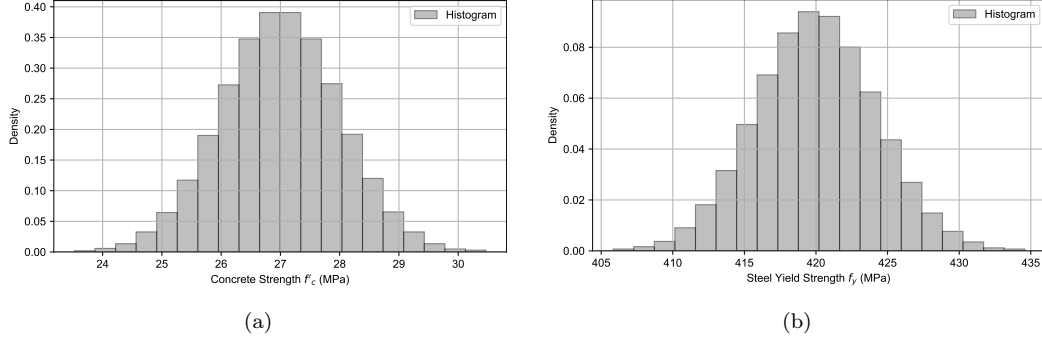


Figure 5: (a) Histogram of concrete compressive strength f'_c (MPa). (b) Histogram of steel yield strength f_y (MPa).

3.4. FE Simulation and Capacity Extraction

Pushover analyses were conducted for each simulation, tracking the force-displacement response at the top of the piers. A total of 3,000 nonlinear pushover simulations were performed, corresponding to Latin Hypercube Sampling (LHS)–generated scour cases across three hydraulic scenarios. For each scour case, the OpenSeesPy model was regenerated to reflect the corresponding scour depth by deactivating `zeroLength` soil springs representing p – y and t – z behavior at or above the scoured-bed elevation across all three pile groups.

After applying and equilibrating gravity loads, a displacement-controlled pushover analysis was performed in the longitudinal direction ($\text{DOF} = 2$) at the deck level, targeting a maximum lateral drift of 5% of the effective bridge height measured from the unscoured riverbed to the deck level. A fallback strategy was implemented to ensure convergence, adaptively switching among Newton–EnergyIncr, Broyden, NewtonLineSearch, and Krylov-based solvers. Each simulation recorded top-node displacements, column rotations, base

430 shears, and moments. The key structural responses at yielding were extracted
 431 and organized into a four-dimensional capacity tuple:

- 432 • *Yield Displacement* (Δ_y): The horizontal displacement at the bilinear
 433 yield point, obtained from curve fitting.
- 434 • *Yield Base Shear* (V_y): The lateral force corresponding to Δ_y .
- 435 • *Yield Moment* (M_x): The bending moment recorded at the base of the
 436 critical columns at yield.
- 437 • *Yield Rotation* (θ_x): The rotational deformation at yielding, computed
 438 from column displacement profiles.

439 This set of yield capacities (Δ_y , V_y , M_x , θ_x) captures essential limit-
 440 state behaviors and supports cross-scenario performance comparisons. Each
 441 pushover response (f vs. d) was smoothed and fitted using a segmented
 442 bilinear function:

$$\hat{f}(d) = \begin{cases} k_1 \cdot d, & d < d_y \\ k_2 \cdot d + (k_1 - k_2) \cdot d_y, & d \geq d_y \end{cases} \quad (7)$$

443 A profile-fitting algorithm (see Listing 1) minimized the squared residuals
 444 between the simulation output f_i and the predicted values $\hat{f}(d_i)$:

$$J(k_1, k_2, d_y) = \sum_{i=1}^N \left[f_i - \hat{f}(d_i; k_1, k_2, d_y) \right]^2 \quad (8)$$

445 The optimal yield displacement d_y was selected by sweeping over a grid
 446 of candidate breakpoints. For each d_y , the best-fit values of k_1 and k_2 were

447 computed using closed-form expressions:

$$k_1 = \max \left\{ 0, \frac{\sum_{i \in I_1} d_i f_i}{\sum_{i \in I_1} d_i^2} \right\}, \quad k_2 = \frac{\sum_{i \in I_2} (d_i - d_y)(f_i - k_1 d_y)}{\sum_{i \in I_2} (d_i - d_y)^2} \quad (9)$$

448 Figure 6 shows an example of bilinear fitting for one simulated scour
 449 depth. The yielding base shear and the corresponding deck-level displace-
 450 ment are extracted from this fit.

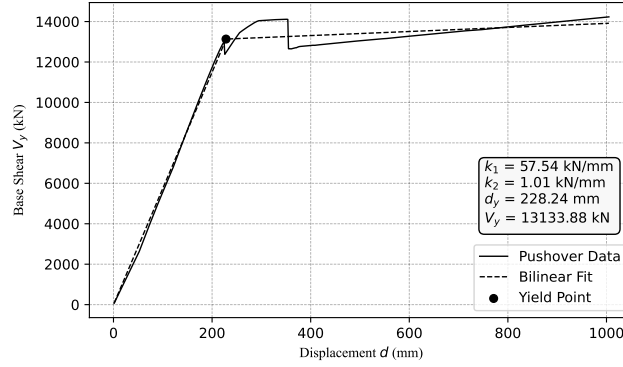


Figure 6: Illustration of capacity-tuple extraction using bilinear pushover curve fitting to extract Δ_y and V_y at a scour depth of 7.347 m.

451 Figure 7 presents histogram plots for the extracted capacity values of all
 452 3,000 simulations, combining the results from all climate scenarios: (a) V_y ,
 453 (b) M_x , (c) Δ_y , and (d) θ_x . A key observation is that the resulting distribu-
 454 tions deviate significantly from common parametric forms—exhibiting both
 455 skewness and multimodality. This indicates that fitting these distributions
 456 to standard probability models (e.g., normal or lognormal) is not meaning-
 457 ful, thereby reinforcing the need for nonparametric uncertainty quantification
 458 methods.

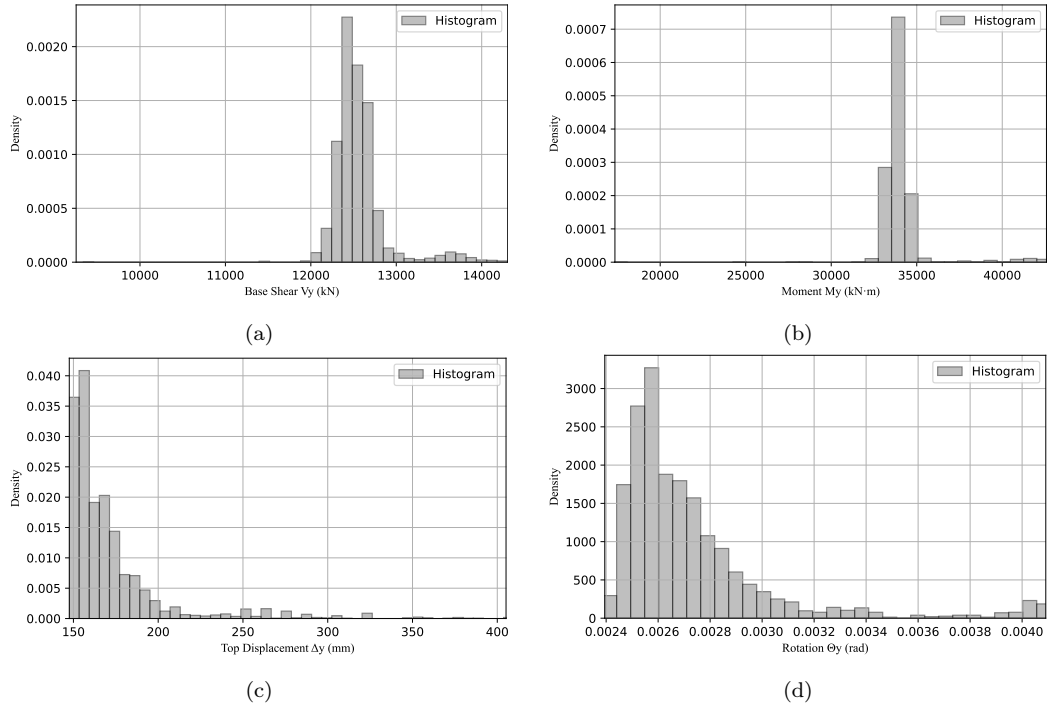


Figure 7: Resulting histogram plots for the bridge capacity tuple: (a) V_y , (b) M_x , (c) Δ_y , and (d) θ_x .

459 3.5. Surrogate Modeling

460 This study develops data-driven surrogate models to predict structural
 461 capacity measures directly from scour depth input. The goal is to replace
 462 computationally intensive finite-element simulations with lightweight predic-
 463 tive models mapping a realized scour depth (s_z) to the four components of
 464 the capacity tuple:

$$s_z \xrightarrow{\text{Predict}} \Delta_y, V_y, M_x, \theta_x \quad (10)$$

465 where Δ_y is the lateral displacement at yielding, V_y is the base shear, M_x is
 466 the flexural moment, and θ_x is the rotational deformation at yield. These sur-
 467rogate models are trained using output from nonlinear pushover simulations,
 468 where each data point links a specific scour condition to a corresponding
 469 capacity tuple.

470 To enrich the input space and improve the model’s ability to approximate
 471 nonlinear patterns, we engineer features based on s_z : s_z^2 , s_z^3 , $\log(s_z)$, $1/s_z$,
 472 and $\sqrt{s_z}$. These transformations help the model capture both polynomial
 473 and asymptotic trends in capacity loss under increasing scour conditions.

474 Before model training, data were cleaned to remove incomplete or unsta-
 475 ble simulations. An outlier filtering step was applied within scour-depth bins
 476 using the interquartile range (IQR) method: points lying outside $[Q1 - 1.5 \times$
 477 $IQR, Q3 + 1.5 \times IQR]$ were discarded (Tukey, 1977). This localized filtering
 478 prevented heavily scoured samples from distorting the statistical characteri-
 479 zation of lightly scoured cases, and vice versa.

480 3.5.1. Support Vector Regression (SVR).

481 Support Vector Regression (SVR) is a kernel-based learning algorithm
 482 that approximates a nonlinear function mapping input features to target
 483 responses. It projects the input vector $\mathbf{x}$ into a high-dimensional space using
 484 a kernel function (here, the Radial Basis Function or RBF kernel) and finds
 485 a function that fits the training data while ignoring small errors within a
 486 specified tolerance band. SVR is especially effective when the training data
 487 are limited and smooth nonlinear relationships exist.

488 Key hyperparameters in SVR include: the penalty parameter C , which
 489 controls the trade-off between model flatness and tolerance to training er-
 490 rors; the ϵ -insensitive zone, which defines the range within which errors are
 491 ignored; and the kernel coefficient γ , which governs the influence of individ-
 492 ual training samples in the feature space. For mathematical formulations
 493 and detailed derivation, see (Smola and Schölkopf, 2004).

Table 2: Combined SVR and GBR hyperparameter settings used in surrogate modeling.

SVR		GBR	
<i>Parameter</i>	<i>Value</i>	<i>Parameter</i>	<i>Value</i>
Kernel	RBF	Number of Estimators	100
ϵ (insensitive zone)	0.01	Learning Rate (ν)	0.05
C (penalty term)	10.0	C_{ub} Tree Depth	3
γ (kernel coefficient)	auto	C_{lb} Samples per Leaf	5
Number of Bootstrap Models	30	Subsample Ratio	0.8
—	—	Number of Bootstrap Models	30

494 3.5.2. Gradient Boosting Regression (GBR).

495 Gradient Boosting Regression (GBR) is an ensemble-based method that
496 builds a predictive model by sequentially adding weak learners, typically
497 shallow regression trees, each of which attempts to correct the residuals of
498 the combined model thus far. This additive process enables the ensemble to
499 capture complex nonlinear relationships in the data.

500 Important GBR hyperparameters include: the number of estimators (i.e.,
501 trees), the learning rate ν which moderates how much each tree contributes to
502 the final prediction, the maximum tree depth which controls model complex-
503 ity, the minimum number of samples per leaf which helps prevent overfitting,
504 and the subsample ratio which governs the fraction of samples used per tree.
505 For mathematical details on gradient boosting, see (Friedman, 2001).

506 3.5.3. Hyperparameters and Performance Metrics

507 Hyperparameter selection for both SVR and GBR was conducted via grid
508 search with 5-fold cross-validation using bootstrapped datasets. Specifically,
509 the optimisation objective was to minimise the mean squared error (MSE)
510 across validation folds. For SVR, the penalty parameter C and kernel pa-
511 rameter γ were varied within logarithmic ranges $[10^{-2}, 10^3]$, while for GBR,
512 the learning rate and maximum tree depth were tuned within $[0.01, 0.2]$ and
513 $[2, 6]$, respectively. Other parameters followed the default settings of the
514 `scikit-learn` implementation. The final hyperparameter settings are pro-
515 vided in Table 2.

516 To assess the predictive performance of the trained surrogate models
517 (SVR and GBR), we employ two widely used evaluation metrics on a held-out
518 test dataset and adopt two classic measures: (1) Coefficient of Determination

519 (R^2), which measures the proportion of variance in the dependent variable
 520 that is predictable from the input variables, and (2) Root Mean Square Error
 521 (RMSE), which is a measure of the average magnitude of prediction errors.
 522 The two metrics together provide a balanced view of both model accuracy
 523 and variance-explaining capability across the different components of the
 524 structural capacity tuple.

525 3.6. Credal-set Uncertainty Quantification

526 A credal set can be defined in its simplest form as an interval prediction: $f(\mathbf{x}) \in$
 527 $[\underline{f}(\mathbf{x}), \bar{f}(\mathbf{x})]$, or more generally as a convex set of distributions $K(\mathbf{x})$ over
 528 the output $f(\mathbf{x})$ given input $\mathbf{x}$, where $K(\mathbf{x})$ denotes the set of all plausible
 529 probability measures consistent with available data and modeling assump-
 530 tions (Walley, 1991). The interval bounds $\underline{f}(\mathbf{x})$ and $\bar{f}(\mathbf{x})$ represent the lower
 531 and upper envelopes of predicted outcomes, capturing epistemic uncertainty
 532 from data scarcity or model ambiguity.

533 In this work, we construct the credal set using bootstrap ensembling
 534 of the surrogate models (SVR and GBR). Specifically, we train $B = 30$
 535 surrogate models per algorithm on independently resampled training sets.
 536 The procedure is:

- 537 • Generate B bootstrap resamples from the original training set (sam-
 538 pling with replacement).
- 539 • Train one surrogate model on each bootstrap sample.
- 540 • For a new input $\mathbf{x}$, obtain B predictions: $\{f_b(\mathbf{x})\}_{b=1}^B$.

541 This ensemble yields an empirical distribution of predicted values at $\mathbf{x}$.
 542 The credal bounds are then defined as:

$$\underline{f}(\mathbf{x}) = \min_{1 \leq b \leq B} f_b(\mathbf{x}), \quad \bar{f}(\mathbf{x}) = \max_{1 \leq b \leq B} f_b(\mathbf{x}) \quad (11)$$

543 Alternatively, one may define bounds based on central quantiles (e.g.,
 544 5th and 95th percentiles) to reduce sensitivity to outliers. The width of the
 545 credal interval reflects the model’s epistemic uncertainty at input $\mathbf{x}$: narrow
 546 intervals suggest higher confidence, while wider intervals indicate sparse or
 547 extrapolative regions of the input space.

548 By adopting this nonparametric, distribution-agnostic formulation, the
 549 bootstrap ensemble enables robust uncertainty quantification under deep
 550 uncertainty, without relying on strong assumptions about the form of the
 551 predictive distribution.

552 4. Results

553 4.1. Surrogate Models and Performance

554 Machine-learning surrogates were trained to predict capacity tuples from
 555 scour depth and derived features, using 80% of the simulated data points
 556 for each capacity and scour pair. Prior to conducting credal-set uncertainty
 557 quantification, model testing was performed using the remaining 20% of data.
 558 Model performance metrics, as shown in Table 3, confirm that GBR outper-
 559 formed SVR across all capacity components, yielding higher R^2 and lower
 560 RMSE. Further explanations are provided below in relation to their predic-
 561 tive behavior.

Table 3: Surrogate Model Performance Metrics (Test Data)

<i>Model</i>	Δ_y		V_y		M_x		θ_x	
	R ²	RMSE	R ²	RMSE	R ²	RMSE	R ²	RMSE
SVR	0.913	0.023	0.894	24.5	0.876	310.2	0.882	0.0017
GBR	0.948	0.018	0.926	20.3	0.910	265.7	0.907	0.0014

562 4.2. Surrogate Prediction and Uncertainty Quantification

563 Figures 8 and 9 present the surrogate ensemble prediction and uncertainty
564 quantification results. Each plot includes (1) the raw simulated data points
565 and (2) the surrogate model predictions with credal-set bounds across scour
566 depths, including the median of the ensemble output, the lower bounds, and
567 the upper bounds. Key observations are summarized as follows.

- 568 • For the simulated capacity–scour data points ($C \in \{\Delta_y, V_y, M_x, \theta_x\}$
569 vs. s_z), the density of data points decreases as s_z increases, reflect-
570 ing the sampling distribution. At extreme scour depths, data points
571 become sparse. However, overall patterns are highly nonlinear:
 - 572 – For strength-related capacities (V_y, M_x), values show slight in-
573 creases at lower scour depths ($s_z < 7.5$ m). Around $s_z = 7.5$ m,
574 M_x exhibits a jump followed by a decline, while V_y shows mild
575 fluctuation before decreasing. Notably, one-to-many mappings or
576 set-valued nonlinearities emerge near $s_z = 7.5$ m, indicating strong
577 nonlinear behaviors in the FE model.
 - 578 – The displacement responses behave differently. The system-level
579 response Δ_y generally increases with scour depth, indicating global

softening. In contrast, the local rotation capacity θ_x increases initially (for $s_z < 7.5$ m), then declines, also displaying non-smooth and set-valued nonlinear characteristics.

- Model inference is represented by the median of the ensemble predictions. The trend lines effectively capture the nonlinear relationships between each capacity and scour depth for all $C \in \{\Delta_y, V_y, M_x, \theta_x\}$. However, SVR and GBR differ in handling non-smooth and set-valued nonlinearities:

- As shown in Figure 8, SVR approximates these nonlinearities through smooth, continuous mappings, as expected from the kernel-based formulation which yields infinitely differentiable predictions.

- As shown in Figure 9, GBR’s predictions are also continuous, but not differentiable. In regions with abrupt nonlinearities, the median predictions exhibit local peaks and discontinuities—an expected consequence of GBR’s piecewise-constant decision-tree architecture.

- The credal-set bounds provide analytical intervals representing epistemic uncertainty in the predicted capacities at each s_z . As seen in Figures 8 and 9, interval widths generally increase with scour depth. Importantly, the upper and lower bounds are asymmetric about the median, as further confirmed in Table 4.

These results indicate that both SVR and GBR effectively approximate the nonlinear relationship between scour depth and structural capacity. GBR

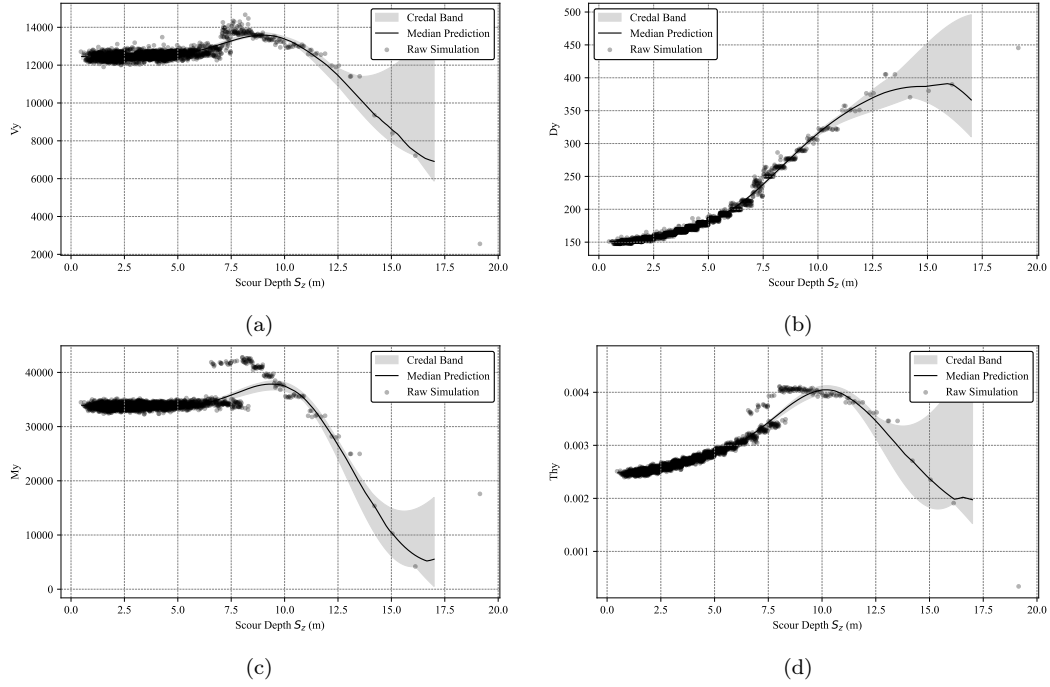


Figure 8: Credal-set surrogate predictions using the SVR models compared against raw simulation data for yield capacities: (a) base shear (V_y), (b) deck displacement (Δ_y), (c) base moment (M_x), and d-rotation (θ_x) as functions of scour depth.

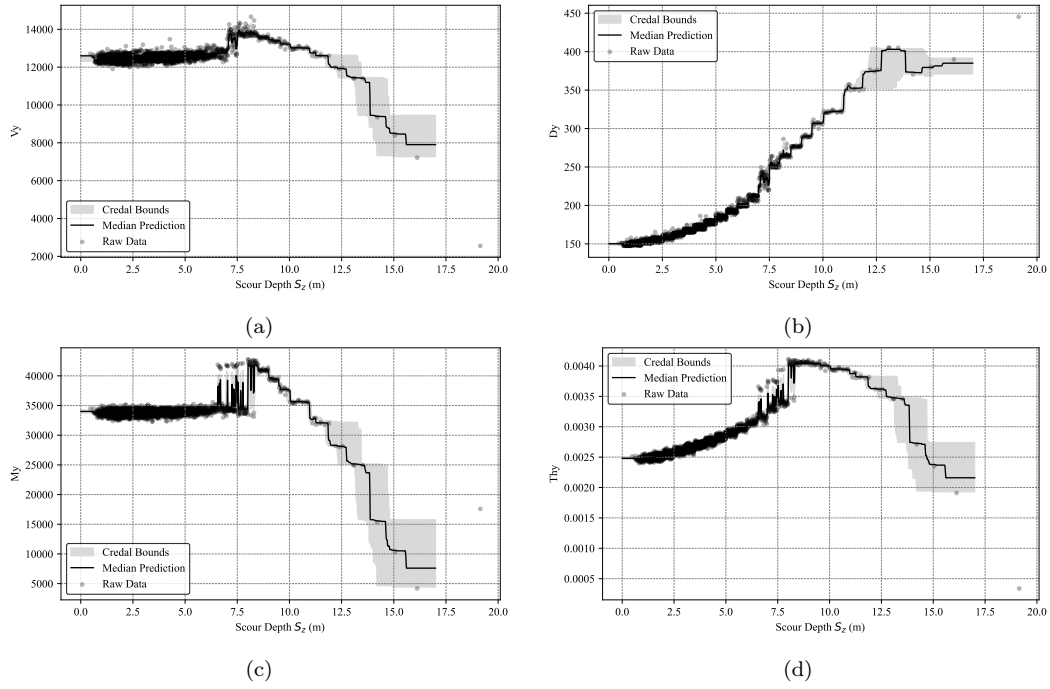


Figure 9: Credal-set surrogate predictions using the GBR models compared against raw simulation data for yield capacities: (a) base shear (V_y), (b) deck displacement (Δ_y), (c) base moment (M_x), and d-rotation (θ_x) as functions of scour depth.

Table 4: Credal-set Bounds for Δ_y and V_y at Key Scour Depths

<i>GBR: Displacement Δ_y (mm)</i>				<i>SVR: Displacement Δ_y (mm)</i>			
<i>Sz (mm)</i>	<i>C_{lb}</i>	<i>C_{med}</i>	<i>C_{ub}</i>	<i>Sz (mm)</i>	<i>C_{lb}</i>	<i>C_{med}</i>	<i>C_{ub}</i>
500	148.78	149.85	150.37	500	164.76	198.47	236.69
1000	148.64	149.44	151.68	1000	148.59	149.69	150.88
5000	174.75	178.46	182.92	5000	181.25	181.51	181.87
10000	305.90	307.51	321.04	10000	309.34	316.61	320.50
15000	371.83	379.42	405.08	15000	272.62	378.04	383.41
<i>GBR: Base Shear V_y (kN)</i>				<i>SVR: Base Shear V_y (kN)</i>			
<i>Sz (mm)</i>	<i>C_{lb}</i>	<i>C_{med}</i>	<i>C_{ub}</i>	<i>Sz (mm)</i>	<i>C_{lb}</i>	<i>C_{med}</i>	<i>C_{ub}</i>
500	12365.31	12444.08	12516.64	500	11257.10	13662.76	15388.41
1000	12328.49	12420.52	12508.12	1000	12332.42	12366.90	12497.20
5000	12311.38	12475.22	12592.03	5000	12582.51	12618.92	12639.30
10000	13016.54	13159.33	13235.98	10000	13050.03	13133.21	13234.92
15000	7346.25	8488.96	11497.43	15000	8313.01	8476.01	11400.96

603 performed better in capturing the capacity transitions when non-smooth and
 604 set-valued relations occur. SVR, using smooth RBF kernels, offered a good
 605 generalization but exhibited limitations in modeling those non-differentiable
 606 relations between the capacity and the scour depth. The credal-set formu-
 607 lation effectively conveys epistemic uncertainty stemming from both model
 608 specification and data sparsity. Prediction intervals widen notably in poorly
 609 sampled regions (i.e., extreme scour), indicating reduced confidence and the
 610 need for conservative estimates in those regimes.

611 *4.3. Feature Dominance Analysis*

612 To interpret surrogate behavior and identify which features most influ-
 613 enced capacity predictions, a feature dominance analysis was conducted for
 614 both SVR and GBR models. The results from both models were largely
 615 consistent (the exception is discussed later). Since the GBR-based interpre-
 616 tation proved more informative due to its superior surrogating performance,
 617 feature dominance results based on the GBR models are described below.

618 Figures 10a to 10d display the GBR-based feature dominance results for
 619 each of the four capacity outputs. These plots illustrate the cumulative con-
 620 tribution of each input feature, where segment count represents the number
 621 of times a feature was selected as a decision-splitting node across all trees
 622 in the ensemble. A higher count indicates a greater influence on predictive
 623 decisions. Figure 11a to 11d visualizes the most dominant feature as the
 624 scour depth varies based on the GBR model as well.

625 The analysis reveals that the logarithmic transformation of scour depth
 626 ($\log Scour$) or $\log(s_z)$, was the most dominant feature across all capacity
 627 components. Other engineered features such as inverse scour depth ($1/s_z$),

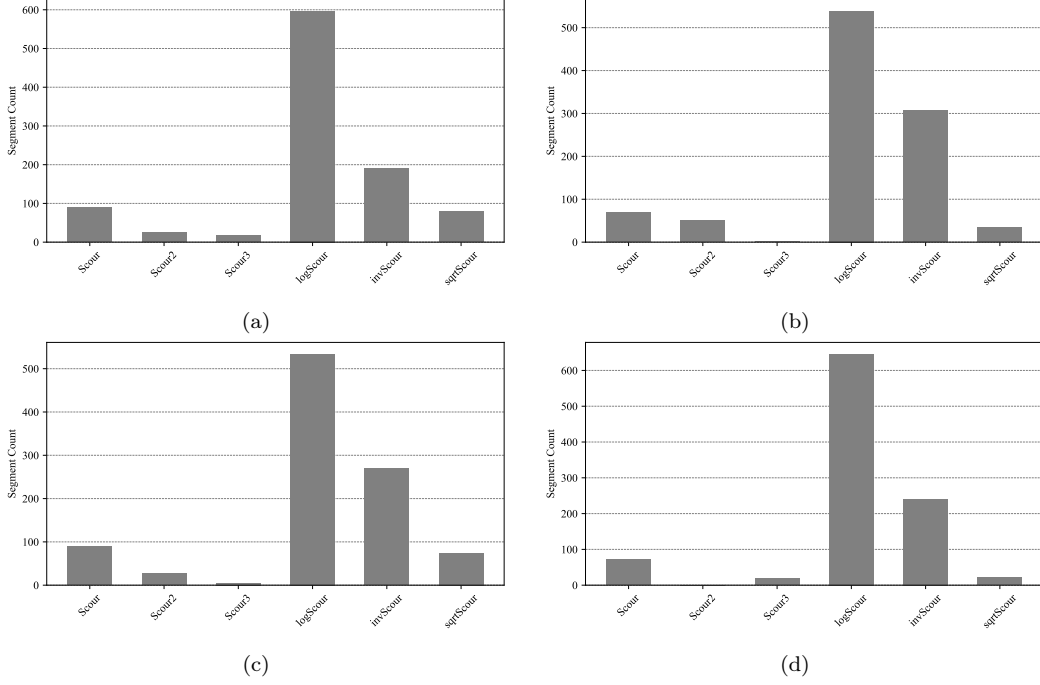


Figure 10: Bar charts of feature dominance counts for each capacity measure, indicating the frequency with which each scour-derived input dominates predictions in the GBR surrogate models: (a) V_y , (b) Δ_y , (c) M_x , and (d) θ_x . Note that the input features of *Scour*, *Scour2*, *Scour3*, *logScour*, *InvScour*, and *sqrtScour* represent s_z , s_z^2 , s_z^3 , $\log(s_z)$, $1/s_z$, and $\sqrt{s_z}$, respectively.

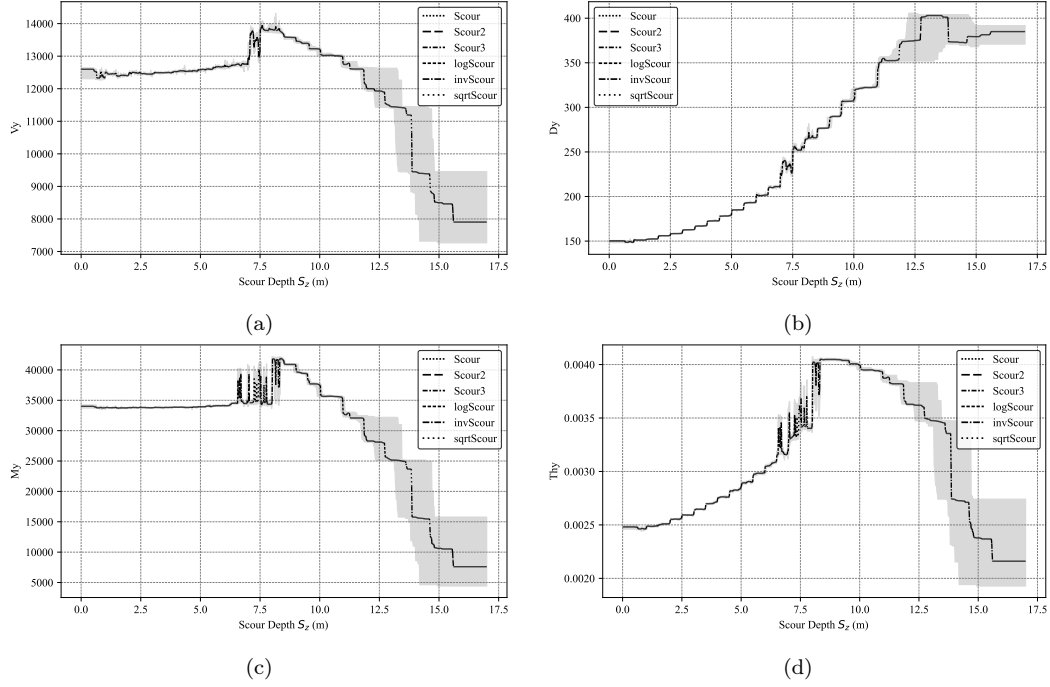


Figure 11: Scour-depth varying visualization of most dominant features for target capacity based on the GBR surrogate models: (a) V_y , (b) Δ_y , (c) M_x , and (d) θ_x . Note that the input features of *Scour*, *Scour2*, *Scour3*, *logScour*, *InvScour*, and *sqrtScour* represent s_z , s_z^2 , s_z^3 , $\log(s_z)$, $1/s_z$, and $\sqrt{s_z}$, respectively.

628 square root of scour depth ($\sqrt{s_z}$), and quadratic terms (s_z^2) gained promi-
629 nence, particularly at larger scour depths, where capacity responses exhibit
630 asymptotic or nonlinear decline. These findings support the inclusion of a
631 diverse set of engineered features to enhance surrogate model expressiveness
632 across different scour regimes.

633 5. Discussion

634 The proposed surrogate modeling framework, integrated with credal-set
635 uncertainty quantification, provides a computationally efficient and conserva-
636 tive approach for evaluating bridge capacity degradation under deep, climate-
637 driven uncertainties. We highlight several limitations, practical implications,
638 and directions for future work, as summarized below.

639 • *Limitation 1: Scope of Uncertainty Representation.* In this work, scour
640 is treated as the dominant causal source of uncertainty when assessing
641 bridge capacity degradation under climate scenarios. Material uncer-
642 tainties (i.e., concrete compressive strength and steel yield strength)
643 were incorporated in the FE simulations as random variables but were
644 treated as background fluctuations rather than causal drivers of scour
645 effects. We recognize that, in a more comprehensive climate-change
646 context, variables such as ambient temperature, moisture, and corro-
647 sion may lead to progressive material degradation and thus partially
648 and causally affect bridge capacity. In such cases, a multi-input sur-
649rogate modeling framework that integrates both environmental and
650 material factors would become essential for accurate and generalizable
651 prediction.

• *Limitation 2: Surrogate Validity and Simulation Fidelity.* Surrogate predictions inherently rely on the fidelity of the FE simulations used for training. Any model misspecification—such as in the soil–structure interaction formulation, boundary condition assumptions, or pier section modeling—may propagate variations into surrogate outcomes. In this work, we observed that non-smooth and set-valued nonlinearities occur when the scour depth reaches about 7.5 m. The resulting nonlinear pushover curve becomes ‘unstable’ in manifesting the yielding points, leading to ‘jumped’ or ‘one-to-many’ localized relations. It is expected that by introducing additional or improved soil springs, such nonlinear behavior may be mitigated, thereby producing stable and more continuous extraction of capacity values. In this work, this is treated as a limitation in the fidelity of the physical system modeling process; nonetheless, it contributes to the epistemic uncertainties. Ultimately, while the surrogates show strong performance on simulated data, the practical deployment of the proposed methodology requires validation of the physical model against experimental, monitored, or field-assessed performance data.

• *Implication 1: Importance of Feature Engineering.* A notable insight is the central role of feature engineering in enhancing model performance. Both SVR and GBR consistently ranked transformed features—particularly $\log(s_z)$ —as most influential. SVR emphasized $\log(s_z)$ and $\sqrt{s_z}$, aligning with its preference for smooth, kernel-based representations. In contrast, GBR favored $\log(s_z)$ and $1/s_z$, reflecting its strength in modeling sharp capacity transitions through decision-tree

677 partitioning.

- 678 • *Implication 2: Extensibility to Capacity–Demand Analysis.* The cur-
679 rent framework focuses on modeling intrinsic structural capacities un-
680 der scour. However, understanding collapse vulnerability necessitates
681 integrating extrinsic response demands, such as those from seismic
682 loading or barge impacts. These demands are inherently stochastic
683 and scenario-dependent. Future work should consider extending the
684 surrogate approach to model both capacities and demands, enabling
685 reliability-based assessment using credal-set formulations that avoid
686 restrictive distributional assumptions.
- 687 • *Practical Value and Future Extension.* The use of surrogate learn-
688 ing and credal-set intervals to frame capacity prediction—namely, in
689 terms of $(C_{lb}, C_{med}, C_{ub})$ rather than a single-valued prediction—offers
690 an informed and interpretable framework for risk-informed decision-
691 making. This is particularly relevant in resilience planning, where
692 high-stakes infrastructure decisions must be made under poorly un-
693 derstood or evolving environmental conditions. Moving forward, inte-
694 grating credal-based predictions into decision models that account for
695 socioeconomic and behavioral factors (e.g., through multi-criteria or
696 game-theoretic frameworks) represents an important avenue for inter-
697 disciplinary extension.

698 6. Conclusion

699 This study developed and demonstrated a surrogate-based framework for
700 evaluating the yield-level capacity degradation of scour-critical bridges un-
701 der deeply uncertain climate-driven scour scenarios. By integrating nonlinear
702 pushover simulations with machine learning surrogates and credal-set uncer-
703 tainty quantification, the framework enables both computational efficiency
704 and distribution-agnostic capacity estimation.

705 Specifically, a suite of nonlinear finite-element analyses was conducted
706 across 3,000 sampled scour conditions, generating a large-scale dataset of
707 capacity responses. These responses were organized as a multi-dimensional
708 capacity tuple (deck displacement, base shear, column moment, and rota-
709 tion) at yield to characterize both superstructure and substructure perfor-
710 mance. Two classes of surrogate models, Support Vector Regression (SVR)
711 and Gradient Boosting Regression (GBR) were trained using both raw and
712 engineered features of scour depth to emulate the simulation outputs. Fea-
713 ture engineering, especially logarithmic and inverse transformations of scour
714 depth, was shown to be critical for capturing nonlinear capacity trends.

715 To address deep epistemic uncertainty associated with future scour sever-
716 ity, a credal-set formulation was employed using bootstrap ensembles. For
717 each surrogate model, bootstrapped resamples were trained and evaluated,
718 yielding lower and upper bounds on capacity predictions for any input scour
719 depth. These credal bounds formed objective and distribution-agnostic in-
720 tervals suitable for decision-making under high uncertainty. Comparative
721 performance evaluation showed that GBR consistently outperformed SVR in
722 prediction accuracy across all capacity metrics, while SVR provided smoother

723 uncertainty envelopes in the scour-depth regime.

724 The results demonstrate that the proposed framework not only reduces
725 computational cost but also explicitly accommodates structural and environ-
726 mental uncertainty in a way that is practical and scalable. The methodology
727 advances current practice by framing capacity prediction as an interval-based
728 decision support tool and offers a path forward for risk-informed infrastruc-
729 ture assessment under progressive climate impacts.

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